

# Alex Siegel

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## EDUCATION

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### University of California, Berkeley

Ph.D. in Biophysics with specialization in Chemical Biology  
Graduate advisor – Prof. David E. Wemmer, Chemistry  
Thesis: Mechanisms of  $\sigma^{54}$  bacterial transcription activation

Berkeley, CA  
June 2016

### California Institute of Technology

B.S. in Biology and Chemistry, double major with honors

Pasadena, CA  
June 2008

## RESEARCH EXPERIENCE

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\*Indicates equal contribution first author publications.

### Postdoctoral Scholar in Chemistry at Caltech

Research with Prof. Shu-ou Shan

Pasadena, CA  
2017-present

- Mapped interactions between a critical plant chaperone, cpSRP43, and membrane protein LHCPs that assemble into photosynthesis machinery by crosslinking-coupled mass spectrometry. ([McAvoy, Siegel, et al, JBC, 2018](#)).
- Identified a new class of cpSRP43 clients involved in chlorophyll synthesis ([Wang et al, PNAS, 2018](#)) and characterized cpSRP43's novel thermoprotection activity towards them ([Ji\\*, Siegel\\* et al, Nature Plants, 2021](#)).
- Discovered a unique activation mechanism triggered by an order-to-disorder transition ([Siegel\\*, McAvoy\\* et al, JMB, 2020](#)) and showed how cpSRP43 can switch from membrane protein LHCPs to globular protein clients ([Siegel et al, In Revision at Science Advances, 2025](#)) or to non-native amyloid-forming clients ([Prov. Patent ID: CIT-9049-P](#)).
- Solved the cryo-EM structure of human mitochondrial ClpB homolog, Skd3, revealing its two roles as a disaggregase chaperone and as a chaperonin-like folding cage ([Gupta\\*, Lentzsch\\*, Siegel\\*, et al, Science Advances, 2023](#)).

### Graduate Researcher in Biophysics at UC Berkeley

Research with Prof. David Wemmer

Berkeley, CA  
2009-2016

- Revealed how NtrC transcriptional activators initiate transcription by threading the N-terminus of an RNA Polymerase subunit through their central pore to apply the force needed to melt DNA ([Siegel et al, JMB, 2016](#)).
- Used molecular tweezers to understand the force-dependent activation of RNA polymerase ([Siegel, dissertation](#)).

## PUBLICATIONS

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\*Indicates equal contribution first author publications.

- **Siegel A**, Kroon G, Zhao CQ, Wang P, Wright PE, and Shan SO. Switchable client specificity in a dual functional chaperone coordinates light harvesting complex biogenesis. In Revision at [Science Advances](#) (2025).
- **Siegel A\***, Singh J\*, Qin PZ & Shan SO EPR Studies of Chaperone Interactions and Dynamics. Biophysics of Molecular Chaperones (eds. Hiller, S., Liu, M. & He, L.) 242–277. [Royal Society of Chemistry](#) (2023).
- Gupta, A\*, Lentzsch AM\*, **Siegel A\***, Yu Z, Chio US, Cheng Y, & Shan SO. Dodecamer assembly of a metazoan AAA+ chaperone couples substrate extraction to refolding. [Science Advances](#). 9(19), eadf5336 (2023).
- Ji S\*, **Siegel A\***, Shan, S., Grimm, B. & Wang, P. Chloroplast SRP43 autonomously protects chlorophyll biosynthesis proteins against heat shock. [Nature Plants](#). 7, 1420-1432 (2021).
- **Siegel A\***, McAvoy C\*, Lam V, Liang FC, Kroon G, Miaou E, Griffin P, Wright P, Shan SO. A Disorder-to-Order Transition Activates an ATP-Independent Membrane Protein Chaperone. [Journal of Molecular Biology](#). 432, 166708 (2020).
- McAvoy C, **Siegel A**, Piskiewicz S, Miaou E, Yu M, Nguyen T, Moradian A, Sweredoski MJ, Hess S, Shan SO. Two Distinct Sites of client protein interaction with the chaperone cpSRP43. [Journal of Biological Chemistry](#). RA118, (2018).

- Wang P, Liang FC, Wittmann D, **Siegel A**, Shan SO, Grimm B. Chloroplast SRP43 acts as a chaperone for glutamyl-tRNA reductase, the rate-limiting enzyme in tetrapyrrole biosynthesis. *Proceedings of the National Academy of Sciences*. 115(15), E3588-96 (2018).
- **Siegel AR**, Wemmer DE. Role of the  $\sigma 54$  Activator Interacting Domain in Bacterial Transcription Initiation. *Journal of Molecular Biology*. 428(23), 4669-85 (2016).
- **Siegel AR**, Mechanisms of  $\sigma 54$  bacterial transcription activation. Doctoral Dissertation University of California, Berkeley. Publication No. 10192026. ProQuest Dissertations Publishing (2016).

## PATENTS

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Patent ID: CIT-9049-P.

Filed in 2023

Provisional patent application for use of chaperones against neurodegenerative diseases.

## PRESENTATIONS

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\*Oral presentation, †Poster presentation ‡Session chair

- **33rd Western Photosynthesis Conference**\*†‡ Biosphere 2, AZ (2024)  
"Regulated capture and delivery of the light harvesting complex proteins by chloroplast SRP."
- **ASBMB annual meeting in Experimental Biology** Spotlight Session\*†‡ Philadelphia, PA (2022)  
"Client Specificity of an ATP-independent Chaperone is Regulated by a Temperature Sensitive Switch"
- **ASBMB Science in a Flash**\* Virtual Meeting (2022)  
"A molecular thermostat that protects plants from heat."
- **Center for Molecular and Cellular Medicine Biochemistry Seminar Series**\* Caltech (2021)  
"A molecular thermostat switches on alternative chaperone activity during heat stress."
- **EMBO workshop: New challenges in protein translocation across membranes**\* Virtual Meeting (2021)  
"A disorder-to-order transition activates the chloroplast signal recognition particle"
- **Center for Molecular and Cellular Medicine Biochemistry Seminar Series**\* Caltech (2021)  
"Regulated substrate capture and release by a membrane protein chaperone"
- **Gordon Research Conference in Computational Aspects of Biomolecular NMR**† Lucca, Italy (2015)  
"Role of the  $\sigma 54$  Activator-Interacting Domain in Bacterial Transcription Initiation"
- **Structure Supergroup Biophysics Seminar Series**\*\* UC Berkeley (2012 & 2014)  
"Mechanisms of  $\sigma 54$  Bacterial Transcription Initiation"
- **Biophysics Annual Retreat**†† UC Berkeley (2010 & 2014)  
"Mechanisms of  $\sigma 54$  Bacterial Transcription Initiation"

## TEACHING

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Courses taught as \*Lead Instructor of Record, †Discussion Section TA, ‡Lab Course TA

- **Advanced Molecular Biology Lab**‡ UC Berkeley (2016)  
Weekly discussion and lab teaching fundamental molecular biology techniques (e.g. cloning, protein expression) and supervising student-written proposals and experimental designs.
- **Biophysics module on protein NMR spectroscopy**\* UC Berkeley (Summer 2012)  
Solo planning and instruction of a 5-week course on practical applications of protein NMR with weekly lectures and hands on collection and analysis of protein NMR spectra.
- **Introductory Chemistry Lab**†‡ UC Berkeley (Summer 2014)  
Taught lab and discussion for 3 sections of 30 students on general chemistry.
- **Biophysical Chemistry**† UC Berkeley (Fall 2010)  
Taught 2 sections of 30 students in weekly discussion sections, and monthly class-wide workshops on biophysics topics, especially kinetics and thermodynamics of macromolecules.
- **Introductory Biology**† Caltech (Spring 2007 & 2008)  
Taught 1 section of 20 undergraduates in two successive years at weekly discussion sections. Developed problems into homework and exams and led monthly review sessions and weekly office hours.

## MENTORING

**Research supervision as primary mentor:** 1 postdoc\*, 7 PhD student†, 6 undergraduate‡, 6 high school§

- Identifying cpSRP43-client binding interfaces by crosslinking-coupled to mass spectrometry\* Sep 2024-present
- Measuring chaperone-client binding by fluorescence lifetime spectroscopy‡ Jun 2024-Aug 2024
- Identifying conformational states of tailor-made chaperones evolved against amyloids† Sep 2023-Dec 2023
- Characterization ligand binding by cpSRP43's two conformational states† Sep 2022-Dec 2022
- Identifying the chaperone-client binding by anisotropy§
- Temperature-dependent conformational changes by circular dichroism and fluorine NMR§ Jun 2021-Jun 2023
- Measuring the thermal dependence of cpSRP43 client binding by FRET§ Jun 2021-Jun 2023
- Developing a split luciferase system to measure chaperone-client binding affinity† Sep 2021-Dec 2021
- Client binding of a superactive chaperone from directed evolution† Sep 2021-Dec 2021
- Measuring ligand induced conformational changes in cpSRP43 by fluorine NMR§§ Aug 2019-Sep 2021
- Monitoring the conformational states of cpSRP43 by fluorine NMR† Feb 2019-Oct 2019
- NMR assignment strategy by site-directed mutagenesis of cpSRP43§§ Sep 2017-Jun 2019
- Monitoring cpSRP43-LHCP client interactions by crosslinking† Jun 2017-Aug 2017
- Force-dependence of the  $\sigma 54$  core binding domain using molecular tweezers† Aug 2013-Dec 2014
- Intrinsic disorder of  $\sigma 54$  domains in the RNA polymerase holoenzyme by NMR† Sep 2014-Dec 2014
- Activation of  $\sigma 54$  by AAA+ ATPase enhancer binding proteins† Jun 2013-Dec 2013
- Dynamics of M. tuberculosis lysine acetyltransferase (Rv0998) by NMR† Sep 2011-Dec 2011
- NMR studies of NtrC4 receiver domain activation‡ Apr 2011-Jun 2012
- Measuring the force-dependent unfolding of the  $\sigma 54$  by molecular tweezers‡ Aug 2010-Jun 2013

## SERVICE

- **CCE Postdoc Committee** 2022-2023  
Liaison between postdoctoral scholars with the Chemistry and Chemical Engineering Division.
- **Caltech Postdoctoral Association (CPA)** 2018-2022  
Secretary/Vice chair working to improve the postdoctoral experience at Caltech. Lobbied for increased funding for the CPA events. Planned dozens of career and social events. Improved onboarding of new postdocs. Coordinated postdocs across all academic divisions.
- **ExploreCaltech** 2019, 2020  
Helped organize ExploreCaltech outreach events. Brought together Caltech scientists across academic divisions to lead interactive booths and give scientific talks for local K-12 students
- **ISEF Judge** 2017  
Judge for biochemistry and biophysics at the Regeneron ISEF.
- **UC Berkeley Graduate Assembly Delegate (GA)** 2011-2012  
Representative from the Biophysics Graduate Group at UC Berkeley responsible for advocating department issues to the graduate assembly
- **Caltech Academic Research Council (ARC)** 2005-2007  
Representative and Secretary of the ARC. Gathered feedback and reported to faculty to make recommendations to improve undergraduate curriculum at Caltech.

## PROFESSIONAL DEVELOPMENT

- **How Learning Works** 2024  
Completion of course for postdocs on evidence-based principles for teaching undergraduates in STEM.
- **Transforming Your Research into Teaching** 2023  
Completion of course for postdocs about how to design and develop college-level courses near our research field.
- **Caltech Certificate of Teaching Interest** 2021-2022  
Completion of a semester-long course and project demonstrating effective teaching. Covering topics including academic research on effective and equitable teaching, effective communication, time management during class, course design using concept mapping, among others.

## SKILLS

\*Experimental or computational skills with proficiency (instruments/software/methods used)

- Independent researcher that can initiate original projects in new areas from gaps in primary literature.
- Scientific communicator to diverse audiences both at conference presentations and through drafting published papers.
- Directly trained over 20 young researchers, including roles as a project leader for group research.
- Lengthy experience with a wide range of techniques in biophysics, biochemistry, and molecular biology (see below).

**NMR Spectroscopy.** Studying protein structure and dynamics by NMR Spectroscopy.

- Isotope labeling (M9 media growth)
- Spectra collection (Bruker Topspin)
- Writing pulse programs (Bruker Topspin)
- Data analysis (NMRpipe, NMRview, CARA, MestreNova)
- Spectrometer maintenance (Bruker 600-900 MHz, nitrogen & helium fills, basic maintenance).
- Site-specific labeling (MTSL & MMTS, <sup>19</sup>F-Phe, BTFA)
- Multiple dimensional collection and analysis (Diffusion ordered spectroscopy, triple resonance assignments, structure determination by NOE)
- Paramagnetic relaxation enhancement
- Non-uniform sampling

**Biophysics Techniques.** Biophysical techniques to characterize protein structure, dynamics, and interactions.

- X-Ray Crystallography
- Mass Spectrometry (Orbitrap Elite, Q-Exactive)
- Electron Paramagnetic Resonance (Bruker CW-EPR)
- Cryo Electron Microscopy (Titan Krios)
- Small Angle X-ray Scattering
- Fluorescence Anisotropy and FRET (Horiba Fluorolog)
- Single molecule FRET (ALEX, Hamamatsu Photonics picosecond streak camera)
- Fluorescence lifetime measurements (Horiba TCSPC)
- Mass Photometry (Refeyn)
- Circular Dichroism (Aviv model 410)
- Molecular Tweezers

**Molecular Biology Techniques.** Cloning, purifying, and *in vitro* assays to study proteins.

- Cloning (site-directed mutagenesis, Gibson Assembly)
- Protein expression (E. coli)
- PCR/qPCR
- Protein purification (FPLC/Chromatography)
- Immunoassays (western blotting and quantification, immunoprecipitation)
- Amber suppression (Schultz method)
- In vitro translation (PURE and self-prepared systems)

**Biochemistry Techniques.** Labeling and other biochemical assays to monitor protein activity.

- Light scattering (Beckman DU series)
- Luciferase folding assays (SpectraMax iD5 plate reader)
- ATPase assays
- Chemical crosslinking (BMOE, BMH, DSSO, DHSO)
- Fluorescence labeling (AlexaFluor/BODIPY maleimides)
- EPR and NMR labeling (MTSL)
- Chromatography (Agilent 1100 series HPLC).

**Software.** Experience with various protein structure packages, bioinformatics tools, and coding languages.

- Structure visualization (ChimeraX, Pymol)
- Cryo-EM analysis (CryoSPARC, Relion)
- Protein structure determination (PHENIX, Coot)
- Mass spectrometry (ProteinProspector, MeroX)
- Structure computation (Gromacs, AlphaFold 2)
- Bioinformatics (BioPython)
- Data analysis (R, Matlab, Graphpad Prism)
- Coding languages (Python)

## REFERENCES

Confidential reference letters are available quickly through Interfolio. Direct contact email also provided.

Prof. **David E. Wemmer**, professor emeritus at UC Berkeley.

Graduate advisor (2009-2016)

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Email: [dewemmer@berkeley.edu](mailto:dewemmer@berkeley.edu)

Prof. **Shu-ou Shan**, Professor at Caltech.

Postdoctoral advisor (2017-present)

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Prof. **Peter Wright**, Professor at The Scripps Research Institute.

Long-term collaborator (2017-present)

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